DS 5007 Natural Language Processing — Assignment 2 Report

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https://github.com/rafiullahkakaredu/i24-7806-NLP-Assignment2

1. Overview

This assignment extends the Assignment-1 BBC Urdu pipeline with neural models written from scratch in PyTorch (no pre-trained models, Gensim, HuggingFace or built-in Transformer modules). The corpus contains 223 articles and 154,678 word tokens from *cleaned.txt*. Articles were labelled with five topics by keyword matching (Politics 29, Sports 37, Economy 22, International 77, Health & Society 58). We build count-based and predictive word embeddings (Part 1), a BiLSTM-CRF for POS tagging and NER on 500 rule-annotated sentences (Part 2) and a Pre-LN Transformer encoder for topic classification (Part 3).

2. Part 1 Results — Word Embeddings

A term–document matrix over the 10,000 most frequent tokens (+<UNK>) was weighted with TF·log(N/(1+df)) (shape 223×7382, vocabulary coverage 100.0%); the top-10 discriminative words per topic are listed in the notebook. A ±5 window co-occurrence matrix gave a PPMI matrix with 461,567 non-zero cells, visualised with t-SNE. Skip-gram with negative sampling (separate V and U matrices, K=10 noise words from $f(w)^{3/4}$, d=100, Adam 10^{-3}, batch 1024) reduced the loss from 4.383 to 2.925 over 10 epochs. 10/10 reported analogies were solved in the top-3 (13/40 of the full pool).

Condition	dim	vocab	MRR (20 pairs)	hits@10
C1 PPMI	7382	7,382	0.214	7
C2 SG raw d=100	100	10,002	0.107	5
C3 SG cleaned d=100	100	7,382	0.138	5
C4 SG cleaned d=200	200	7,382	0.180	5

Table 1. Four-condition embedding comparison (MRR over 20 manually labelled word pairs).

Best condition by MRR: C1 PPMI (MRR=0.214). Skip-gram on cleaned text (C3, 0.138) vs raw text (C2, 0.107): cleaning helps because normalisation merges spelling variants and strips attached punctuation, so counts are not split across surface forms. PPMI (C1, 0.214) gives sparse, high-dimensional vectors that are strong for frequent words but noisy for rare ones. Doubling d to 200 (C4, 0.180) improves the result: the larger space separates more fine-grained relations, although the gain should be weighed against twice the memory and training cost.

3. Part 2 Results — POS Tagging and NER

500 sentences (Health & Society 167, International 167, Sports 166) were annotated by a rule-based tagger (Assignment-1 stemmer/lemmatiser + lexicons of 361 nouns, 1715 verb forms, 319 adjectives, 253 adverbs; 12 tags incl. AUX) and a gazetteer (90 Pakistani persons, 117 locations, 82 organisations) in BIO format, then split 350/75/75 stratified by topic. The 2-layer BiLSTM (forward and backward states concatenated, dropout 0.5, packed sequences, padding masked) was trained with Adam and early stopping (patience 5). POS test accuracy is **93.76%** with macro-F1 **91.35** (fine-tuned embeddings); the most confused pairs were NOUN<->ADJ (46), NOUN<->VERB (33), NOUN<->ADV (10).

Embeddings	POS val macro-F1	POS test acc.	POS test macro-F1	NER val F1
frozen	88.32	91.24	88.20	75.00
fine-tuned	92.19	93.76	91.35	78.46

Table 2. Frozen vs. fine-tuned Word2Vec (C3) initialisation.

Entity	P (CRF)	R (CRF)	F1 (CRF)	F1 (softmax)	support
PER	85.71	66.67	75.00	50.00	9
LOC	87.80	83.72	85.71	77.65	43
ORG	64.29	69.23	66.67	76.92	13
MISC	50.00	30.00	37.50	0.00	10
overall	79.41	72.00	75.52	69.17	75

Table 3. NER test results (conlleval, entity level): BiLSTM-CRF with Viterbi vs. softmax decoding.

Run	POS test macro-F1	NER test F1
Baseline	91.35	75.52
A1 Unidirectional LSTM	88.81	63.64
A2 No dropout	93.86	80.00
A3 Random embedding init	93.04	70.23
A4 Softmax instead of CRF (NER)	–	69.17

Table 4. Ablations A1–A4 (one change at a time, same split).

A1 Unidirectional LSTM: POS macro-F1 -2.54, NER F1 -11.88 vs. baseline: removing backward context hurts, so it contributes to the baseline. A2 No dropout: POS macro-F1 +2.51, NER F1 +4.48 vs. baseline: removing dropout regularisation actually helps here (+7.0 combined); with only ~350 silver-labelled training sentences the model benefits from more capacity / less regularisation. A3 Random embedding init: POS macro-F1 +1.69, NER F1 -5.29 vs. baseline: removing pre-trained Word2Vec initialisation hurts, so it contributes to the baseline. A4 Softmax instead of CRF: NER F1 -6.35: structured CRF decoding helps by forbidding invalid BIO transitions (e.g. O→I-PER) and keeping multi-word entities together.

Error analysis on the test set (14 false positives, 21 false negatives; 5 of each are explained in the notebook) groups the errors as: boundary/partial (9), never seen (9), context generalisation (not in gazetteer) (7), type confusion (5), ambiguous context (5). Context-generalisation false positives are often plausible entities absent from the silver gazetteer, i.e. the model generalises beyond its labels.

4. Part 3 Results — Transformer Encoder

Articles were truncated/padded to 256 tokens and split 156/33/34 (stratified). The encoder uses separate scaled dot-product attention, 4-head self-attention (d_model=128, d_k=d_v=32, per-head projections), FFN (512, ReLU), a fixed sinusoidal positional buffer, 4 Pre-LN blocks and a learned [CLS] token with a 128→64→5 MLP, trained with AdamW ($5 \cdot 10^{-4}$, wd 0.01) and cosine schedule with 50 warm-up steps for 20 epochs. Test accuracy is **58.82%** and macro-F1 **50.47**; a BiLSTM classifier on the same split reaches 67.65% / 62.53.

Model	test acc. (%)	macro-F1	best val epoch	epochs to 95%	s / epoch	params
BiLSTM	67.65	62.53	18	12	7.83	1,385,757
Transformer	58.82	50.47	12	12	5.24	1,746,949

Table 5. BiLSTM vs. Transformer on 5-class topic classification.

(1) On the held-out test set the BiLSTM is more accurate: 67.6% vs. 58.8% for the Transformer, a difference of 8.8 percentage points (macro-F1 50.5 for the Transformer vs. 62.5 for the BiLSTM). (2) Both models reached 95% of their best validation accuracy after 12

epoch(s). (3) Per epoch the Transformer was faster (5.24s vs. 7.83s). (4) The final-layer heatmaps show that the heads specialise: head 1 is the most focused (mean maximum weight 0.02 per query), head 3 the most diffuse (0.01), and on average 0.6% of each token's attention goes to the [CLS] position. (5) For a corpus of only 200–300 articles the BiLSTM is the more appropriate architecture, which this run also supports.

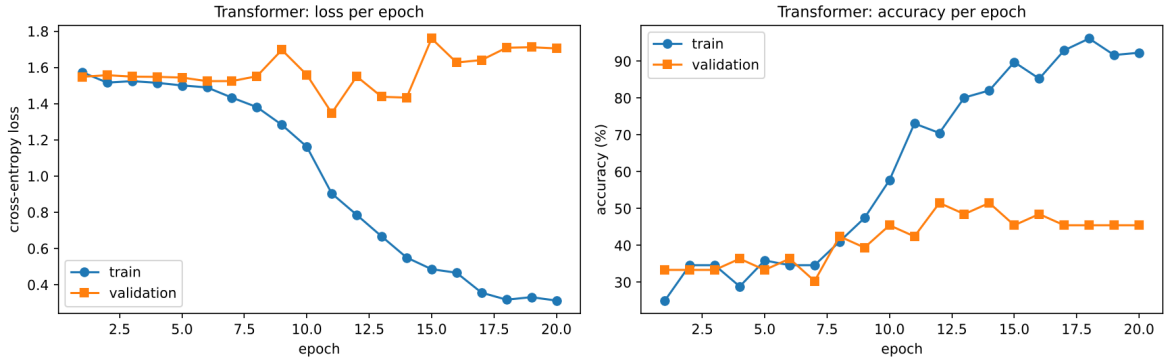


Figure 1. Transformer training/validation loss and accuracy per epoch.

5. Conclusion

All three parts were implemented from scratch and run end-to-end. Skip-gram embeddings trained on cleaned text (C1 PPMI best by MRR) capture topical and morphological regularities and transfer usefully to sequence labelling. The BiLSTM-CRF reaches 75.5 entity F1 and 93.8% POS accuracy on silver labels, with bidirectionality, dropout and pre-trained initialisation evaluated by ablation. On the small topic corpus the Transformer obtains 58.8% accuracy; with only a few hundred articles a recurrent model with pre-trained embeddings remains the safer choice. Main limitations are the silver (rule-generated) labels and the small corpus; manual correction of annotations and more articles would be the next steps.